

Annex 7

WHO Biowaiver List: proposal to waive in vivo bioequivalence requirements for WHO Model List of Essential Medicines immediate-release, solid oral dosage forms

Update of WHO Biowaiver List: proposal to waive in vivo bioequivalence requirements for WHO Model List of Essential Medicines immediate-release, solid oral dosage forms, WHO Technical Report Series No. 1060, 2025, Annex 5.

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1. Introduction and background

The World Health Organization (WHO) recognizes the possibility of waiving in vivo bioequivalence studies for immediate-release, solid oral dosage forms with active pharmaceutical ingredients (APIs) belonging to classes I and III according to the Biopharmaceutical Classification System (BCS), using comparative dissolution studies as surrogate proof of bioequivalence ([1](#)).

The WHO solubility classification, also referred to as the WHO Biowaiver List, is a tool for national regulatory authorities and pharmaceutical manufacturing companies, suggesting medical products that are eligible for a waiver from in vivo bioequivalence studies, which are usually necessary to establish the therapeutic equivalence with the originator (comparator). For exemption from an in vivo bioequivalence study, an immediate-release, multisource (generic) product should exhibit very rapid or rapid in vitro dissolution characteristics that are comparable to those of the reference product. A risk-based evaluation should also account for the excipients used in the formulation of the finished pharmaceutical product.

In addition, the present list replaces the existing literature-based compilation published in 2006 that is reported in the *Proposal to waive in vivo bioequivalence requirements for WHO Model List of Essential Medicines immediate-release, solid oral dosage forms* ([2](#)) based on data extracted from the public domain (that is, solubility data published by different authors using inconsistent experimental conditions).

The WHO Biowaiver Project is organized into study cycles. Previous and current cycles are summarized below in order to provide an outline of the project development:

- 2018: cycle I, also referred to as the pilot phase.
- 2019: cycle II
- 2020: cycle III
- 2021: cycle IV
- 2022/2023: cycle V
- 2024: cycle VI
- 2025: cycle VII
- 2026/2027: cycle VIII – current study cycle.

2. WHO solubility classification for biowaiver

In 2017, the Fifty-second Expert Committee on Specifications for Pharmaceutical Preparations (ECSPP) recommended that the WHO Secretariat revise the existing list using verifiable laboratory data that are generated according to consistent WHO

criteria. Acting on this directive from the ECSPP, the WHO Secretariat initiated a multicentre research project, the Biowaiver Project, aimed at experimentally determining the equilibrium solubility profile of medicines listed in the WHO Model List of Essential Medicines, using a harmonized approach (3).

To classify APIs according to the BCS framework, two critical properties are usually evaluated: (a) an API's aqueous solubility; and (b) its absorption or permeability. The initial phase of the WHO Biowaiver Project centres on unambiguous experimental assessment of the solubility parameter, as only highly soluble APIs are eligible for biowaiver. Once experimental solubility data are available, the exact BCS class assignment can be determined by utilizing quantitative absorption or permeability data. However, since high solubility within an aqueous environment is a necessary prerequisite for an API to be eligible for a waiver from bioequivalence studies, the current focus on solubility is justified to guide the regulatory decision.

The WHO classification should be considered a living document and is meant to be regularly updated in accordance with new quality requirements and progress in scientific development.

3. Scope

The aim of the WHO Biowaiver List is to enable an informed decision as to whether or not a waiver from in vivo bioequivalence studies could be granted safely according to the *WHO guideline on Biopharmaceutics Classification System-based biowaivers* (1).

The WHO Biowaiver List is expected to promote access to standard-quality essential medicines by shortening the time required to develop a multisource (generic) product supporting an optimized pharmaceutical development.

The WHO Biowaiver List has been recognized by WHO regional and country offices as a “global good” – a normative work essential to strengthening global health in WHO Member States.

4. Methodology

The *WHO Protocol to conduct equilibrium solubility experiments for the purpose of biopharmaceutics classification system-based classification of active pharmaceutical ingredients for biowaiver* (4) is a tool available to all participants in this research. It was developed with the purpose of providing a harmonized methodology for the equilibrium solubility experiments, thereby minimizing the potential source of variability among centres and studies.

APIs studied were received primarily as in-kind donations from pharmaceutical manufacturers supporting WHO in this scientific work.

Equilibrium solubility experiments were conducted by universities, official national control laboratories and WHO collaborating centres.

5. Results

[Table A7.1](#) provides an overview of the APIs studied by WHO during cycles I to VII. The new APIs studied in cycle VII are reported in bold. Fixed-dose combination products, where all APIs contained in the combination drug product were studied as monocomponents ([Table A7.1](#)), are also reported in [Table A7.2](#).

Table A7.1

WHO solubility classification of active pharmaceutical ingredients prioritized from the WHO Model List of Essential Medicines

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
abacavir (sulfate)	Antiretrovirals	Antiretrovirals (HIV)	600	Yes	I/III
aciclovir	Antiviral medicines	Antiherpes medicines	800	No	II/IV*
allopurinol	Medicines used to treat gout, supportive medicines	Chronic gout, hyperuricaemia	900	No	II/IV*
amitriptyline (hydrochloride)	Medicines for mental and behavioural disorders	Medicines used in depressive disorders	75	No	I/III
amlodipine (besylate)	Cardiovascular medicines	Antihypertensive medicines	10	No	I/III
amoxicillin (trihydrate)	Antibacterials	Antibiotics	3000	Yes	II/IV*
apixaban	Anticoagulant	Prevention of stroke, deep vein thrombosis and pulmonary embolism	10	No	II/IV**
azithromycin (dihydrate)	Antibacterials	Antibiotics	2000	Yes	II/IV
benznidazole	Antiprotozoal medicines	Chagas disease	400	No	II/IV*

Table A7.1 *continued*

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
biperiden (hydrochloride)	Anticholinergic medicines	Parkinson disease	2	No	I/III
bisoprolol (fumarate)	Cardiovascular medicines	Antihypertensive medicines	20	No	I/III**
cefalexin (monohydrate)	Antibacterials	Antibiotics (access group)	2000	No	I/III**
cefixime (trihydrate)	Antibacterials	Antibiotics	400	No	II/IV
chloroquine phosphate	Antiprotozoal medicines	Antimalarial medicines	1000 mg salt (= 600 mg base)	No	I/III
ciprofloxacin hydrochloride	Antibacterials	Antibiotics (watch group)	750	No	II/ IV*
clindamycin (hydrochloride)	Antibacterials	Access group antibiotics	450	Yes	I/III
codeine (phosphate hemihydrate)	Medicines for pain and palliative care	Opioid analgesics	60	No	I/III
cycloserine	Antibacterials	Antituberculosis medicines	1000	Yes	I/III
daclatasvir (dihydrochloride)	Antiviral medicines	Medicines for hepatitis C	60	Yes	II/IV**
darunavir (ethanolate)	Antiviral medicines	Antiretrovirals (HIV)	800	Yes	II/IV**

Table A7.1 *continued*

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
dexamethasone	1. Gastrointestinal medicines 2. Immunomodulators and antineoplastics 3. Medicines for pain and palliative care	1. Antiemetic medicines 2. Acute lymphoblastic leukaemia, multiple myeloma 3. Medicines for other common symptoms in palliative care	1, 3: 0.5 to 10 mg a day, depending on the disease being treated 2: 40 mg	Yes	I/III**
dolutegravir	Antiviral medicines	Antiretrovirals (HIV)	50	Yes	II/IV**
doxycycline (hyclate)	1. Antiprotozoals 2. Antibacterials	1. Antimalarial medicines 2. Antibiotics (access group)	100	No	I/III**
efavirenz	Antiviral medicines	Antiretrovirals (HIV)	600	Yes	II/IV
emtricitabine	Antiviral medicines	Antiretrovirals (HIV)	200	Yes	I/III**
entecavir	Antiviral medicines	Antihepatitis medicines	1	Yes	I/III**
ethambutol (hydrochloride)	Antibacterials	Antituberculosis medicines	2000	Yes	I/III
ethionamide	Antibacterials	Antituberculosis medicines	500–1000	Yes	II/IV*

Table A7.1 *continued*

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
fluconazole (form III)	Antifungal medicines	Cryptococcosis and candidosis	800	Yes	I/III
furosemide	Cardiovascular medicines	Medicines used in heart failure	80	No	II/IV
hydralazine (hydrochloride)	Cardiovascular medicines	Antihypertensive medicines (pregnancy-induced hypertension)	100	No	I/III
hydrochlorothiazide	Cardiovascular medicines	1. Medicines used in heart failure 2. Antihypertensive medicines 3. Diuretics	100	No	I/III
hydroxychloroquine (sulfate)	Disease-modifying anti-rheumatic drugs (DMARDs)	Lupus erythematosus	600	No	I/III**
isoniazid	Antibacterials	Antituberculosis medicines	300	Yes	I/III
ivermectin	Anti-infective medicines	Anthelmintic medicines, scabies	40	Yes	II/IV
lamivudine	Antiviral medicines	Antiretrovirals (HIV)	300	Yes	I/III

Table A7.1 *continued*

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
levofloxacin hemihydrate	Antibacterials	Antituberculosis medicines	1500	Yes	I/III
levonorgestrel	Medicines for reproductive health and perinatal care	Oral hormonal contraceptives	1.5	Yes	II/IV*
mefloquine (hydrochloride)	Antiprotozoal medicines	Antimalarial medicines	1250 (as hydrochloride)	Yes	II/IV
methyl dopa (sesquidrate)	Cardiovascular medicines	Pregnancy-induced hypertension	500	No	I/III
miltefosine	Antiprotozoal medicines	Antileishmaniasis medicines	50	Yes	I/III**
misoprostol (as 1% dispersion in hydroxypropyl methylcellulose)	Uterotonic medicines	Postpartum haemorrhage	0.8	Yes	I/III**
moxifloxacin hydrochloride	Antibacterials	Antituberculosis medicines	400	Yes	I/III**
ondansetron (base)	Antiemetic medicines, medicines for pain and palliative care	Nausea or vomiting, palliative care	24	No	II/IV**
oseltamivir (phosphate)	Antiviral medicines	Influenza virus	75 (as phosphate)	Yes	I/III**

Table A7.1 *continued*

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
paracetamol	Medicines for pain and palliative care, antimigraine medicines	Non-opioids and non-steroidal anti-inflammatory medicines, treatment of acute attack	1000	No	I/III
primaquine (phosphate)	Antiprotozoal medicines	Antimalarial medicines (curative treatment of <i>Plasmodium vivax</i> and <i>P. ovale</i> infections)	15	Yes	I/III
proguanil (hydrochloride)	Antiprotozoal medicines	Antimalarial medicines	200	No	I/III
pyrazinamide	Antibacterials	Antituberculosis medicines	2000	Yes	I/III
pyrimethamine	Antiprotozoal medicines	Antimalarial medicines	75	Yes	II/IV
quinine (sulfate)	Antiprotozoal medicines	Antimalarial medicines	648	No	II/IV*
raltegravir (potassium)	Antiviral medicines	Antiretrovirals (HIV in pregnant women and in second line)	400	Yes	II/IV**

Table A7.1 *continued*

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
ribavirin	Antiviral medicines	Viral haemorrhagic fevers	600	Yes	I/III**
rifampicin	Antibacterials	Antituberculosis, antileprosy medicines	750	Yes	II/IV
sofosbuvir	Antiviral medicines	Medicines for hepatitis C	400	Yes	II/IV**
tenofovir disoproxil (fumarate)	Antiviral medicines	Antiretrovirals (HIV)	300	Yes	I/III**
valganciclovir	Antiviral medicines	Cytomegalovirus retinitis (CMVr)	900	Yes	I/III**
zidovudine	Antiviral medicines	Antiretrovirals (HIV)	300	Yes	I/III

API: active pharmaceutical ingredient; PQ: prequalification; EOI-PQ: expression of interest for prequalification.

Note: In the table, the new APIs studied in cycle VII are reported in bold text.

^a 23rd WHO Model List of Essential Medicines (2023) (3).

^b According to the *WHO guideline on Biopharmaceutics Classification System-based biowaivers* (1), APIs belonging to classes I and III are eligible for biowaiver. Once experimental permeability data are available, the exact class attribution will be possible (that is, either class I or class III). The present solubility characterization is already sufficient to provide an indication as to whether or not an API is eligible for biowaiver.

* Change in solubility class with respect to WHO 2006 classification.

** APIs characterized for the first time within the WHO Biowaiver Project.

Table A7.2

WHO solubility classification of fixed-dose combination products prioritized from the WHO Model List of Essential Medicines

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
efavirenz + emtricitabine + tenofovir disoproxil (fumarate)	Antiviral medicines	Antiretrovirals (HIV)	600 + 200 + 300	Yes	II/IV**
efavirenz + lamivudine + tenofovir disoproxil (fumarate)	Antiviral medicines	Antiretrovirals (HIV)	600 + 300 + 300	Yes	II/IV**
emtricitabine + tenofovir disoproxil (fumarate)	Antiviral medicines	Antiretrovirals (HIV)	200 + 300	Yes	I/III**
ethambutol + isoniazid + pyrazinamide + rifampicin	Antibacterials	Antituberculosis medicines	2000 + 300 + 2000 + 750	Yes	II/IV
ethambutol + isoniazid + rifampicin	Antibacterials	Antituberculosis medicines	2000 + 300 + 750	Yes	II/IV**
isoniazid + pyrazinamide + rifampicin	Antibacterials	Antituberculosis medicines	300 + 2000 + 750	Yes	II/IV

Table A7.2 *continued*

Medicine ^a	Therapeutic area	Indication	Highest therapeutic dose (mg)	API PQ, EOI-PQ	WHO classification ^b
lamivudine + zidovudine	Antiviral medicines	Antiretrovirals (HIV)	300 + 300	Yes	I/III**

API: active pharmaceutical ingredient; PQ: prequalification; EOI-PQ: expression of interest for prequalification.

Note: In the table, the new fixed-dose combination products studied in cycle VII are reported in **bold text**.

^a 23rd WHO Model List of Essential Medicines (2023) (3).

^b According to the *WHO guideline on Biopharmaceutics Classification System-based biowaivers* (1), APIs belonging to classes I and III are eligible for biowaiver. Once experimental permeability data are available, the exact class attribution will be possible (that is, either class I or class III). The present solubility characterization is already sufficient to provide an indication as to whether or not an API is eligible for biowaiver.

* Change in solubility class with respect to WHO 2006 classification.

** APIs characterized for the first time within the WHO Biowaiver Project.

Establishing a new WHO Biowaiver List that is based on unambiguous verifiable experimental solubility data is a critical project with tremendous public health implications for patients, procurers, United Nations agencies, national and regional regulatory authorities, payers, ethics committees and manufacturers worldwide. The involvement and support of WHO stakeholders and partners is highly encouraged and appreciated.

References

1. WHO guideline on Biopharmaceutics Classification System-based biowaivers. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-seventh report (Annex 7). Geneva: World Health Organization; 2024 (WHO Technical Report Series, No. 1052; <https://iris.who.int/handle/10665/376607>). Licence: CC BY-NC-SA 3.0 IGO.
2. Proposal to waive in vivo bioequivalence requirements for WHO Model List of Essential Medicines immediate-release, solid oral dosage forms. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fortieth report (Annex 8). Geneva: World Health Organization; 2006 (WHO Technical Report Series, No. 937; <https://iris.who.int/handle/10665/43443>).
3. The selection and use of essential medicines 2023: web annex A: World Health Organization Model List of Essential Medicines: 23rd list (2023). Geneva: World Health Organization; 2023 (<https://iris.who.int/handle/10665/371090>). Licence: CC BY-NC-SA 3.0 IGO.
4. Protocol to conduct equilibrium solubility experiments for the purpose of Biopharmaceutics Classification System-based classification of active pharmaceutical ingredients for biowaiver. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-third report (Annex 4). Geneva: World Health Organization; 2019 (WHO Technical Report Series, No. 1019; <https://iris.who.int/handle/10665/312316>). Licence: CC BY-NC-SA 3.0 IGO.

Further reading

- Guidance on the selection of comparator pharmaceutical products for equivalence assessment of interchangeable multisource (generic) products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: forty-ninth report (Annex 8). Geneva: World Health Organization; 2015 (WHO Technical Report Series, No. 992; <https://iris.who.int/handle/10665/176954>).
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- General background notes and list of international comparator pharmaceutical products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-first report (Annex 5). Geneva: World Health Organization; 2017 (WHO Technical Report Series, No. 1003; <https://iris.who.int/handle/10665/258720>). Licence: CC BY-NC-SA 3.0 IGO.
- Multisource (generic) pharmaceutical products: guidelines on registration requirements to establish interchangeability. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-seventh report (Annex 8). Geneva: World Health Organization; 2024 (WHO Technical Report Series, No. 1052; <https://iris.who.int/handle/10665/376607>). Licence: CC BY-NC-SA 3.0 IGO.
- WHO list of international comparator products. Geneva: World Health Organization; 2024 (<https://iris.who.int/handle/10665/376309>). Licence: CC BY-NC-SA 3.0 IGO.